

A Wood Densitometer Using Direct Scanning with X-Rays

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Summary. A computerised wood densitometer system using X-rays is described. The technique uses 5 mm increment core samples, machined to 2 mm thickness in the axial direction of the fibre, and an isotopic radiation source (Fe 55). Direct measurement of the intensity of radiation passing through the sample enables wood density variations to be recorded automatically and stored on disc file for subsequent analyses with an interactive computer program.

Introduction

Radiation densitometry is a commonly used technique for assessing density characteristics of wood samples. Beta-ray and X-ray methods were pioneered in the U.K. (Phillips 1960) and France (Polge 1963). The X-ray system gained widespread acceptance in principle (Parker and Kennedy 1973) and by 1980 there were some 30 laboratories in 18 countries engaged in densitometry (Evertsen 1980). Most of these institutions use variations of the Polge method, i.e., production of a negative image of a wood sample by exposure of the wood and a film to X-rays generated electrically. The films must be developed under carefully controlled conditions and analysed in some form of optical densitometer to give analog or digital output of wood density variations. A detailed description of such a system is given by Parker et al. (1980).

Wood Densitometry in New Zealand

Wood densitometry has been used as a research tool at the New Zealand Forest Research Institute (FRI) since 1968. An operational beta-ray system was developed

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The wood densitometer components were designed and built by staff of the Institute of Nuclear Sciences Laboratory, DSIR, Lower Hutt. Interfacing of the wood densitometer components was carried out by R. Thomas, Forest Research Institute, Rotorua

and described by Harris (1969). Since that time modifications have been made to the data collection and analysis procedures, culminating in 1979 with the replacement of the original equipment with a new densitometer, custom-built by the Institute of Nuclear Sciences Laboratory of the DSIR. The method adopted was similar to that described by Woods and Lawhon (1974) using an isotopic source of X-rays (Fe 55) but included complete computer control of the scanning operation.

A feature of the beta-ray method was the direct scanning of wood samples prepared from 10 mm diameter increment cores. There were several advantages to this approach, viz.

1. Low levels of radiation were used, obviating the need for a special lead-lined chamber.

2. No photographic processes were involved, thus avoiding several potential sources of error (e.g., random variations in film density and development characteristics).

Disadvantages included:

1. The large size of the wood sample necessitated laborious field collection procedures.

2. Slow scanning speed (2 mm/min).

3. The relatively large beam size (2.0 mm $\times$ 0.5 mm) gave poor density resolution in growth rings less than 5 mm width.

4. The analog output required tedious manual interpretation.

The new X-ray method retains the above advantages while overcoming most of the disadvantages by: (a) permitting the use of 5 mm diameter increment core samples, (b) reducing the beam size, and (c) increasing the feed speed.

There were several strong reasons for preferring an isotopic source, i.e., cost (no special cabinets and no maintenance), simplicity of operation, and stable medium-term (in minutes) and short-term (in milliseconds) output. The short half life results in a decrease in the scanning speed over time, and it is recommended that the source be replaced approximately every two years.

The Direct-Scanning Densitometer System

Wood samples for densitometry (5 mm diameter increment cores or radial blocks machined to 5 mm $\times$ 5 mm) are resin-extracted by refluxing with methanol in a Soxhlet apparatus for 24 hours, then conditioned to 10% moisture content by storing in a controlled atmosphere (25 °C and 50% relative humidity). Distortion of the wood is prevented through the use of drying racks. A twin-blade saw (New Zealand Forest Service 1977) reduces the samples to 2 mm thick in the longitudinal direction. The 75 mm diameter saw blades must be sharpened regularly to maintain a high-quality finish on the transverse plane of the wood (Fig. 1).

The machined samples are mounted in the moveable densitometer carriage in preparation for scanning.

The wood densitometer system (Figs. 2–4) comprises:

1. The Fe 55 radiation source, collimated to 2.0 $\times$ 0.1 mm, emitting 6 KeV radiation,

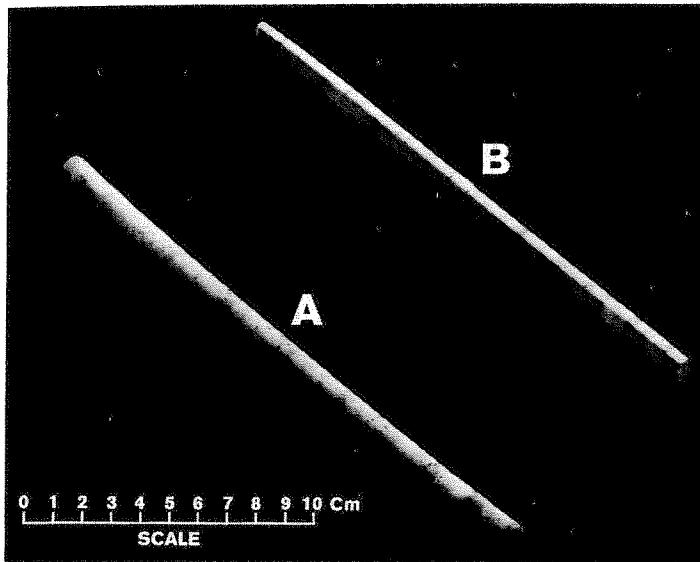


Fig. 1. Increment core samples: A before cutting, B after cutting

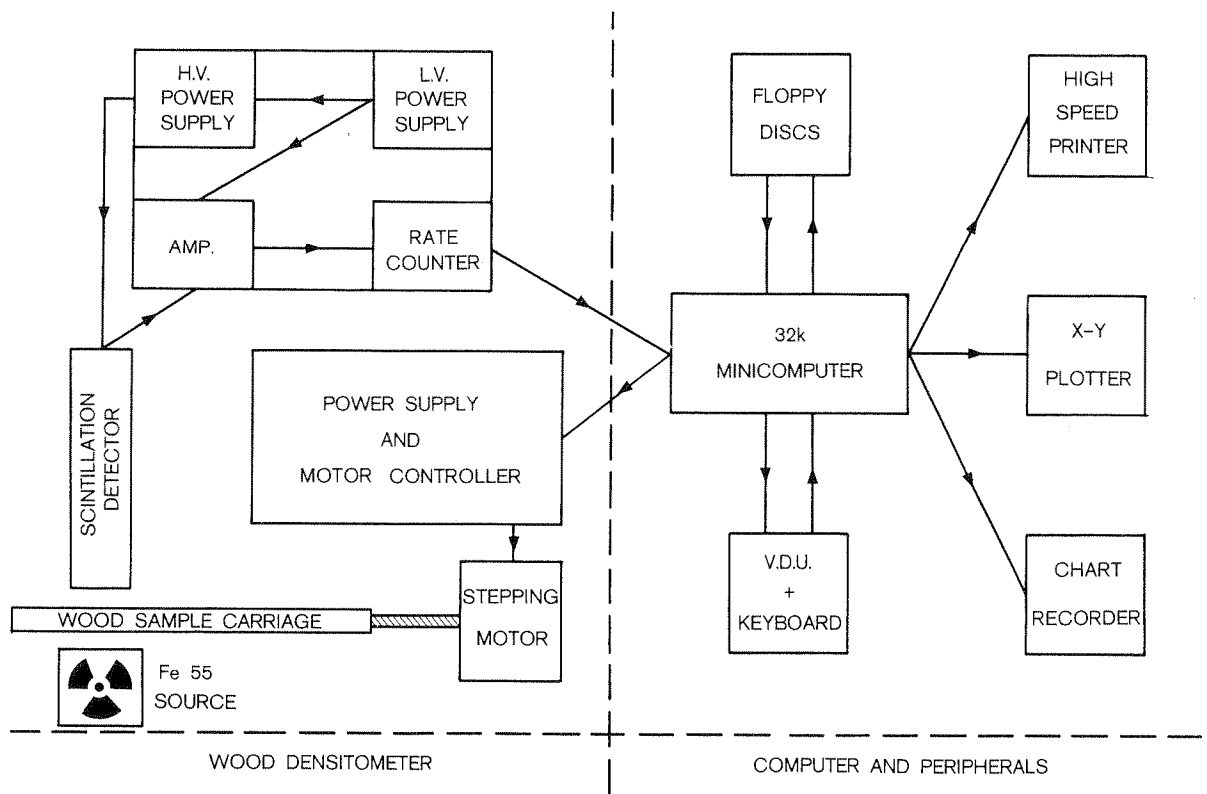


Fig. 2. Schematic diagram of wood densitometer system

2. the sample carriage, driven by a stepping motor in multiples of 0.0125 mm,
3. the X-ray scintillation detector,
4. the pulse counter, and
5. the on-line computer and peripheral devices (keyboard, VDU, line printer, chartrecorder, printer/plotter and twin floppy disc units).



Fig. 3. Overview of wood densitometer system, showing: 1 wood densitometer, 2 computer, 3 visual display and keyboard, 4 chartrecorder, 5 line printer, and 6 printer/plotter

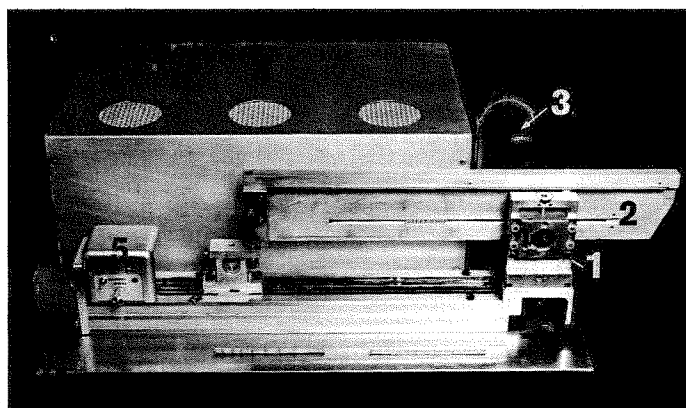


Fig. 4. Wood densitometer, showing: 1 Fe 55 radiation source, 2 sample carriage, 3 scintillation detector, 4 power supply, and 5 stepping motor

Data Acquisition

The scanning procedure is controlled by the keyboard operator, using computer programs. Two phases are involved initially, viz. calibration and scanning.

Calibration (Program CALB)

The degree of attenuation of the collimated beam when passing through wood depends on: a) the sample thickness, b) sample moisture content, c) sample specific gravity, and d) the attenuation coefficient for wood.

Slight variations in (a) and (b) may occur despite rigid quality control measures, so it is recommended that an attenuation factor be derived for each batch of wood samples. This factor will automatically compensate for variations in (a), (b), and (d).

The attenuation factor is calculated using a number of short (e.g., 20 mm) lengths of wood from the batch to be scanned. Basic density is determined by the standard maximum moisture content method (Smith, 1954) and the subsample is machined at the same time as the main batch. The prepared calibration strips are scanned in the densitometer using program CALB which determines the average radiation count rate in air and the average rate for 0.1 mm increments along each sample. The mean attenuation factor is then calculated from the expression (adapted from Meredith and Massey 1972):

$$\text{Attenuation Factor (A.F.)} = \text{Log}_e \left(\frac{C_0}{C_1} \right) \times \frac{1}{D}, \text{ where}$$

C_0 Average count rate in air,

C_1 Average count rate in wood,

D Basic density of the wood.

The expression, $D = \text{Log}_e \left(\frac{C_0}{C_1} \right) \times \frac{1}{\text{A.F.}}$ is subsequently used in the scanning program to convert count rates to wood densities.

Scanning (Program SCANNA)

The strength of the isotopic source diminishes perceptibly with time (half life 2.7 years) and so it is necessary to re-evaluate the air count rate periodically. This is an optional first step in the SCANNA program and is accomplished by measuring the time (in milliseconds) required to accumulate a count rate of 4000 pulses. An average of 400 such assessments is used to give an estimate of the mean source strength.

Prior to the actual scanning process, the keyboard operator defines:

1. The number of steps to be incremented by the stepping motor between density readings (normally 8 ($\times 0.0125$ mm) to correspond with the 0.1 mm beam width). Density resolution and scanning speed can be altered by changing this parameter.
2. The appropriate attenuation factor (from program CALB).
3. The chartrecorder option (yes or no).
4. The number of critical pulses/density reading, which determines when the stepping motor is advanced (normally 300). This controls the scanning speed and ensures that each channel is assessed to the same degree of precision. Operational scanning speeds range between 10 and 20 mm/minute.
5. Sample identification.

When these parameters have been set, control of the scanning passes to the computer, which advances the sample carriage until the wood intercepts the X-ray beam. The densitometer controller then activates the pulse counter at 10 ms intervals until the critical count has been exceeded. The actual count and the number of intervals taken are used to calculate the density which is stored on a floppy disc file and the carriage is advanced (e.g. 0.1 mm) by the stepping motor. Scanning continues until the air count rate is again detected, at which point the operator is given the option either to continue scanning the same sample (as in the

case of discontinuous strips) or loading a new sample and recommencing the procedure. On completion of each scan, the disc file contains a record of the scanning parameters and the channel sample in sequence from the centre of the tree outwards.

Data processing (Program EDITOR)

The density data acquired by the SCANNA program consists of a one-dimensional array which must be processed to give information on individual growth ring characteristics. This is achieved using program EDITOR. EDITOR reads the raw data from the disc files and performs two main functions:

1. Identifies growth ring boundaries by detecting the abrupt latewood/early-wood transition.
2. Allocates growth ring numbers and years to each ring, using information supplied by the keyboard operator (number from pith, if known, and year of first ring scanned).

Before a hard copy of the data is produced, there is a very important screening phase during which the operator may interactively modify the results. Windows of edited data, 80 channels in width, are displayed on the VDU screen (Fig. 5) and the operator can use a number of command options, viz.

1. Ring boundaries can be deleted if they do not correspond to annual growth rings, for example in the case of false latewood rings.
2. Ring boundaries can be inserted where density resolution was insufficient to trigger a boundary identification automatically.
3. A range of channels may be ignored in later ring width and density calculations, or values may be "smoothed" between two specified channels. These options are used when there are discontinuities or abnormalities in the sample (cracks, resin canals, resin pockets, etc).
4. The window may be positioned at any point along the sample.
5. The window may be centred on any specific growth ring.
6. The entire sample may be viewed sequentially by drawing successive windows with a short pause between each.

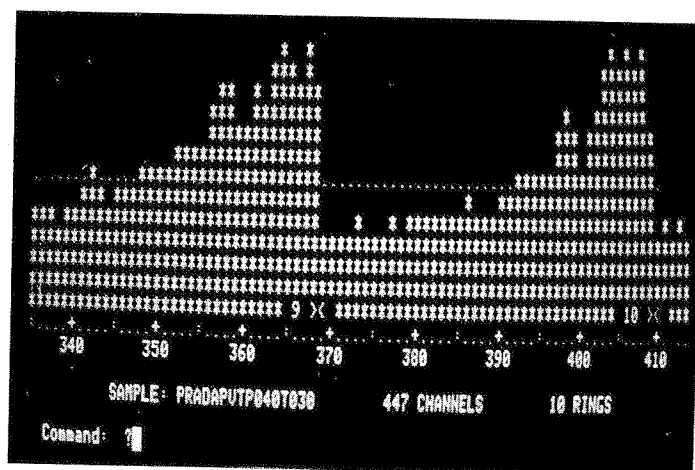


Fig. 5. Visual display of 80 scanned channels

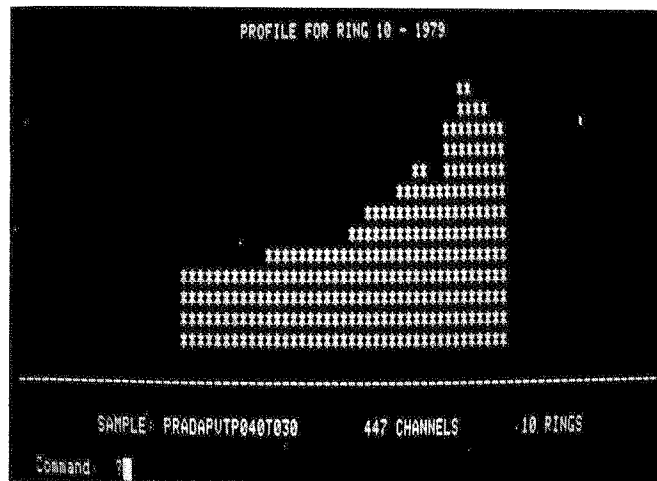


Fig. 6. Visual display of summarised single ring profile

A terminating command causes EDITOR to analyse each growth ring in sequence and calculate:

- Ring width components – earlywood, latewood and whole ring widths. The earlywood/latewood density transition is nominated by the operator and is usually 500 kg/m^3 .
- Ring density components – earlywood, latewood, minimum, maximum and ring mean densities. Also calculated are ring uniformity (latewood – earlywood) and range (maximum – minimum).
- A standardised ring profile for each annual ring, comprising 20 channels (Fig. 6).

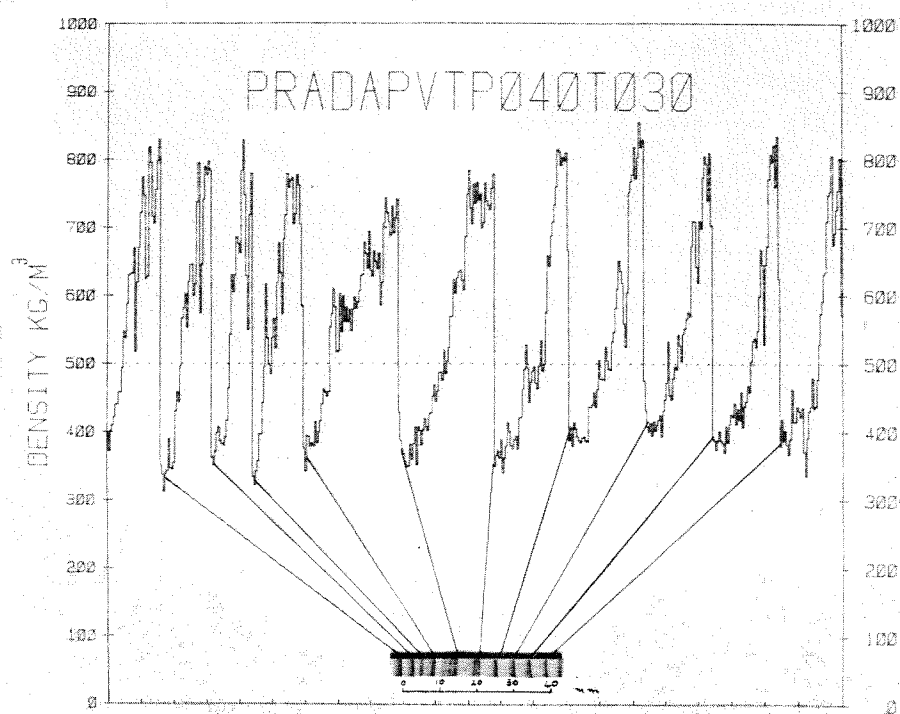


Fig. 7. Printer/plotter record of scanned core

Table 1. Data summary produced by program EDITOR

Ring from pith	Year	Ring width components (mm)				Ring density components (kg/m ³)						Max.	Range
		Outer radius	Ring width	Early wood	Late wood	% Late wood	Ring mean	Early wood	Late wood	Unif.	Min.		
1	1961	8.3	8.3	8.3	0	0	308	308	0	0	167	438	271
2	1962	31.7	23.4	23.4	0	0	337	337	0	0	262	452	190
3	1963	47.7	16.0	16.0	0	0	328	328	0	0	237	451	214
4	1964	70.2	22.5	22.2	.3	1	346	344	517	173	247	531	284
5	1965	80.0	9.8	9.1	.7	7	349	336	516	180	259	566	307
6	1966	90.4	10.4	9.1	1.3	12	400	381	534	153	227	590	363
7	1967	99.6	9.2	6.9	2.3	25	406	351	574	223	263	646	383
8	1968	104.9	5.3	3.4	1.9	35	419	326	585	259	236	666	430
9	1969	112.0	7.1	3.5	3.6	50	476	340	608	268	272	713	441
10	1970	122.3	10.3	5.9	4.4	42	463	379	575	196	291	651	360
11	1971	130.9	8.6	4.3	4.3	50	493	373	612	239	302	697	395
12	1972	142.8	11.9	4.5	7.4	62	529	437	586	149	379	737	358
13	1973	153.6	10.8	6.9	3.9	36	462	387	595	208	317	713	396
14	1974	162.6	9.0	4.3	4.7	52	538	397	606	269	322	822	500
15	1975	172.8	10.2	4.3	5.9	57	513	424	578	154	357	702	345
16	1976	180.3	7.5	5.1	2.4	32	458	407	567	160	360	669	309
17	1977	193.6	13.3	4.5	8.8	66	567	400	653	253	315	845	530
18	1978	210.1	16.5	6.3	10.2	61	536	409	615	206	349	739	390
19	1979	217.6	7.5	3.5	4.0	53	538	414	646	232	345	763	418
20	1980	221.8	4.2	1.6	2.6	61	551	392	649	257	318	789	471

Core length = 221.8 mm
Core density = 437 kg/m³
X-sectional density = 484 kg/m³
EW/SW transition = 500 kg/m³
ID: Core PR542

All these summarised data are stored on disc file and a hard copy of the ring width and density components produced on the line printer (Table 1). A visual record of the sample density may be obtained from the printer/plotter (Fig. 7).

A limitation of the method currently in use is the short half-life of the Fe 55 source, which will necessitate replacement at two-year intervals to ensure an adequate scan speed. A recent publication (Jacoby and Perry 1981) indicates that a small X-ray generator may have an application in this situation and this is being investigated. Nevertheless, the current system has proved its worth in a number of studies, e.g., an assessment of thinning and fertiliser effects on *Pinus radiata* (Cown and McConchie 1981).

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